

Data Quality Report

Customer Segmentation & Sales Intelligence Platform

Version: 1.0

Prepared By: Rizwan Hussain

Status: Final

Data Source: Online Retail II Dataset

1. Data Quality Assessment

The initial dataset contained **1,067,371 transaction-level records across 8 columns**. The profiling process identified missing values, duplicate records, missing customer identifiers, cancelled transactions, and invalid pricing records.

These issues were assessed based on their potential impact on customer-level analysis, revenue calculations, and customer segmentation.

2. Initial Data Quality Findings

Issue	Records
Total Records	1,067,371
Missing Values	247,389
Duplicate Records	34,335
Missing Customer IDs	235,151
Cancelled Invoices	18,390*
Invalid Quantities	0*
Invalid Prices	70*

*These counts represent records removed during the sequential cleaning process after previous cleaning steps had already been applied.

3. Data Cleaning Decisions

Data Quality Issue	Action Taken	Business Justification
Duplicate Records	Removed	Prevent duplicate transactions from inflating sales and customer activity.
Missing Customer IDs	Removed	Customer ID is required for customer-level analysis and RFM segmentation.
Cancelled Invoices	Removed	Cancelled transactions do not represent completed purchases.
Non-positive Quantities	Removed	Customer segmentation focuses on completed purchases.
Non-positive Prices	Removed	Invalid prices would distort revenue calculations.
Missing Values	Resolved through applicable cleaning steps	Ensures the analytical dataset contains complete records required for segmentation.

4. Cleaning Audit

Stage	Rows Remaining	Rows Removed
Original Dataset	1,067,371	—
After Removing Duplicates	1,033,036	34,335
After Removing Missing Customer IDs	797,885	235,151
After Removing Cancelled Invoices	779,495	18,390
After Removing Invalid Quantities	779,495	0
After Removing Invalid Prices	779,425	70

5. Final Dataset Quality

After cleaning, the dataset contains:

- **779,425 transaction records**
- **10 analytical columns**
- **5,878 unique customers**
- **4,631 unique products**
- **36,969 unique invoices**
- **41 countries**
- **17,374,804.27 total transaction revenue**
- **0 missing values**
- **0 duplicate records**
- **0 invalid quantities**
- **0 invalid prices**

6. Final Assessment

The cleaned dataset is considered suitable for customer-level behavioral analysis, RFM feature engineering, customer segmentation, SQL analysis, and Power BI reporting. Cleaning decisions were made based on the analytical requirements of the project and the need to prevent invalid or non-completed transactions from distorting customer and revenue metrics.
